CTMT Seismic Morphism:
Global Catalogue Falsification and a Local Declustered Physical Chart

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Abstract

We report a two-level seismic test of the current CTMT morphism-family formulation. First, a global earthquake catalogue is used as an honest falsification attempt: annual event-feature charts are built from real catalogue data, rolling three-year transport monodromy loops are computed, and the CTMT diagnostic family is evaluated. Second, a local declustered seismic chart is introduced using a Gardner–Knopoff style mainshock/aftershock separation. The local chart is not a large statistical validation; it is a physically interpretable chart showing how CTMT morphism diagnostics should be anchored when dependent aftershock structure is explicitly removed.

The combined result supports a conservative conclusion: CTMT exhibits nontrivial internal structure. Its diagnostics are neither redundant nor arbitrary, and the framework has become more constrained in response to counterexamples rather than more permissive. This suggests that CTMT is capturing a genuine mathematical organization of transport diagnostics, while the ultimate physical scope remains open.

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1 Purpose

The purpose of this paper is not earthquake prediction. The purpose is to test whether the final CTMT morphism statement remains meaningful when confronted with real seismic catalogues and with a physically meaningful local declustering operation.

The working CTMT morphism statement is

$$\mathrm{Mor}_{\mathrm{CTMT}}(\mathcal{X}) = \{f : f^*\Phi = \Phi \text{ for all } \Phi \in \mathcal{I}_{\mathrm{CTMT}}\}. \tag{1.1}$$

This definition is intentionally not scalar. It says that an admissible CTMT morphism is a structure-preserving map class over a family of transport observables.

Remark 1 (Interpretive constraint)**.** The correct conclusion from the current evidence is not that CTMT has found one universal seismic invariant. The correct conclusion is that the invariants form a constrained hierarchy: operator monodromy, fixed-sector persistence, phase/winding behavior, self-reference, rank stability, and coarse-graining stability detect different failure modes.

2 Seismic background: why declustering matters

Seismic catalogues contain dependent events such as foreshocks and aftershocks in addition to independent mainshock or background events. Khurram and Khalid describe catalogue declustering as a process used to remove dependent events such as foreshocks and aftershocks and to separate mainshocks in distance–time windows. Their Pakistan catalogue contains 34112 events, and their declustering found 2714 clusters with 19512 clustered events, corresponding to 57.19% of the catalogue [1].

The same paper explicitly uses Gardner–Knopoff style time and distance windows and reports representative thresholds such as $d = 89$ km and $t = 173$ days for $M = 6.5$, $d = 100$ km and $t = 322$ days for $M = 7.0$, and $d = 254$ km and $t = 980$ days for $M = 8.0$ [1]. The catalogue study also reports a magnitude of completeness $M_c = 3.8$ for the full region and a Gutenberg–Richter b-value near 0.78 in the maximum-likelihood solution [1].

For CTMT, declustering is not merely a preprocessing choice. It identifies the object that is transported. A raw catalogue mixes independent forcing structure with dependent relaxation structure. A declustered chart separates these sectors.

3 Global seismic falsification attempt

3.1 Global data and limitations

The uploaded global earthquake catalogue has 782 parsed rows in the requested 2001–2023 filter, but the actual uploaded file covers 2001–2022. The maximum magnitude in the uploaded file is 9.1, corresponding to the 2011 Tohoku and 2004 Sumatra–Andaman events. The Kaggle dataset page describes the same column semantics, including magnitude, time, CDI, MMI, alert, tsunami flag, significance, network, stations, distance, gap, magnitude type, depth, latitude, longitude, location, continent, and country [3].

3.2 Annual seismic charts

Each event is embedded into a feature vector containing:

- magnitude and log-energy proxy;
- depth;
- CDI and MMI;

- significance, tsunami flag, and alert score;
- spherical geographic coordinates.

Features are robustly standardized. For each year y, a local annual seismic chart B_y is obtained from the leading principal frame of that year's event cloud.

3.3 Transport monodromy

For adjacent annual charts define

$$T_{ab} = \arg\min_T \left(\|B_b T - B_a\|_F^2 + \lambda \|T\|_F^2 \right). \tag{3.1}$$

For a rolling three-year loop,

$$H_y = T_{y+2,y} T_{y+1,y+2} T_{y,y+1}. \tag{3.2}$$

The diagnostic family is

$$\mathcal{I}_{\mathrm{CTMT}} = \{I_H, I_F, I_W, I_S, I_R, I_C\}, \tag{3.3}$$

where I_H is operator loop defect, I_F fixed-sector persistence, I_W phase/winding proxy, I_S self-reference closure, I_R rank/seepage stability, and I_C coarse-graining or quotient-style stability.

3.4 Global result

The global test produced both stable and defective annual loops. The result is not reducible to magnitude or impact severity alone. In the generated run, background-like, intermediate, and high-energy/impact regimes all contain morphism defects, and high-energy/impact loops are not automatically all defective.

Regime	Loops	Defect rate	Operator defect	Winding defect	Coarse quotient defect
background-like	6	0.667	0.167	0.167	0.167
high-energy/impact	6	0.500	0.000	0.000	0.000
intermediate	8	0.625	0.375	0.000	0.125

Empirical finding 1 (Global catalogue falsification state)**.** The global catalogue run does not collapse CTMT diagnostics into a single scalar. Operator loop defect, phase variation, self-reference, rank stability, and quotient stability respond differently across annual regimes. Therefore the real-data test supports the morphism-family interpretation operationally, while remaining insufficient as a physical seismic law.

4 Local declustered physical chart

4.1 Data supplied

The local declustered example contains six events. The type label M denotes an independent mainshock surviving the time–distance window filter, whereas A denotes a dependent aftershock removed by the declustering rule.

Event	Latitude	Longitude	Depth	Magnitude	Type	d_w km	t_w days
MX2009_001	28.983	-113.015	10.0	6.2	M	54.0	79.0
MX2009_002	28.891	-112.980	12.3	4.8	A	30.5	18.0
MX2009_003	29.012	-113.110	9.1	5.0	A	32.0	22.0
MX2009_004	28.950	-112.890	11.5	4.1	A	25.0	11.5
MX2009_005	29.120	-113.250	8.0	4.3	A	26.5	13.0
MX2009_006	31.250	-115.420	15.0	5.8	M	48.0	61.0

The local chart contains two independent mainshocks and four dependent aftershocks. Thus the mainshock fraction is $2/6 = 0.333$ and the aftershock fraction is $4/6 = 0.667$. The local time span is approximately 89.4 hours.

4.2 Declustering as a CTMT projection

Let the raw event cloud be

$$\mathcal{D}_{raw} = \mathcal{M} \cup \mathcal{A}, \tag{4.1}$$

where $\mathcal{M}$ is the independent mainshock sector and $\mathcal{A}$ is the dependent aftershock sector. The declustering map is a physical projection

$$\Pi_M : \mathcal{D}_{raw} \to \mathcal{M}. \tag{4.2}$$

Definition 1 (Declustered seismic chart)**.** A declustered seismic chart is a triple

$$\mathfrak{S} = (\mathcal{M}, \mathcal{A}, \Pi_M), \tag{4.3}$$

where $\mathcal{M}$ is the independent mainshock sector, $\mathcal{A}$ is the dependent aftershock sector, and Π_M is a time–distance window projection that preserves mainshock events and removes dependent events.

Proposition 1 (Local physical chart constraint)**.** *A CTMT morphism on a declustered seismic chart is not admissible merely because the morphism preserves raw event-cloud statistics. The morphism must either preserve the mainshock projection* Π_M *or explicitly declare that the morphism acts on the dependent aftershock response sector.*

Proof 2. The local data contain a raw event cloud with two independent and four dependent events. A map can preserve raw-cloud summaries while changing the mainshock skeleton by moving, merging, or deleting one of the two M events. Conversely, a map can preserve the mainshock skeleton while discarding dependent aftershock multiplicity. Therefore raw-cloud preservation and mainshock-sector preservation are distinct morphism constraints.

5 Jalisco physical chart context

The Jalisco region is a physically meaningful next test case, because the region sits at the interaction of the North America, Cocos, and Rivera plates and forms the Jalisco block. The RESAJ paper states that the region is among the most tectonically active in Mexico, includes large historical earthquakes, tsunamis, and active volcanoes, and that RESAJ was initiated to improve seismic-hazard study and seismic-parameter characterization for building-code design [2].

The same paper reports that RESAJ currently has 28 stations, with 26 telemetered in real time and two autonomous stations, and that the station distribution allows improved study of Bahía de Banderas–Islas Marías, the northern Jalisco coast and seismic gap, Colima and Tepic–Zacoalco grabens, intraplate seismicity within the Jalisco block, and volcano seismicity. The paper also reports that the station distribution permits automatic detection at approximately $M \geq 2$ across the region and $M \geq 1$ in some areas [2].

For CTMT, this is precisely the missing local physical chart. A serious test should not only use global annual PCA, but should use regional tectonic windows, local station completeness, local detection thresholds, and declustered mainshock/aftershock sectors.

6 Combined interpretation

The global and local tests play different roles:

1. The global test is an honest falsification attempt: it asks whether CTMT diagnostics collapse into one scalar or merely track magnitude. They do not.
2. The local declustered chart introduces the missing physical projection: the seismic chart must specify whether it transports raw catalogue structure, independent mainshock structure, or dependent aftershock response structure.
3. The Jalisco/RESAJ context identifies the correct physical scale for a stronger test: regional tectonic windows with local station/network metadata.

Theorem 1 (Current seismic morphism state)**.** *On the present evidence, CTMT seismic morphism is not finalized by raw catalogue monodromy alone. A defensible seismic morphism must preserve a family of transport diagnostics and must specify whether the transported seismic object is raw catalogue structure, independent mainshock structure, or dependent aftershock response structure.*

Remark 2 (Conservative interpretation)**.** CTMT exhibits nontrivial internal structure. Its diagnostics are neither redundant nor arbitrary, and the theory has become more constrained in response to counterexamples rather than more permissive. This strongly suggests that the framework is capturing a genuine mathematical organization, even though its ultimate scope and physical significance remain open questions.

7 Non-claims

This paper does not claim earthquake prediction, a causal tectonic law, universal CTMT morphism finality, or that six local declustered rows are sufficient for statistical validation. The local declustered chart demonstrates the correct physical projection needed for a serious seismic chart. A publishable seismic validation requires a larger regional declustered catalogue, station/network metadata, and independent replication.

8 Next experiment

The next CTMT seismic experiment should use a regional catalogue with the following sectors:

- raw event cloud;
- Gardner–Knopoff mainshock projection;
- dependent aftershock response sector;
- local station/network completeness sector;
- regional tectonic segmentation;
- window-refinement and coarse-graining tests.

For each sector, CTMT should compute the same transport-observable family and test whether morphism defects are stable under declustering, coarse-graining, and regional window refinement.

A Python appendix: local declustered seismic chart

Conceptual principle. CTMT does not introduce an arbitrary morphism. The morphism is reconstructed from the observable response of measurements under admissible perturbations of the transported object.

Once the transported object (here: the declustered seismic chart) and admissible perturbations (here: anchor position and declustering scales) are fixed, the forward map $F(\theta)$ is uniquely induced by how observable aftershock coordinates respond to those perturbations.

Companion file. The companion file `ctmt_local_seismic_chart_appendix.py` implements the local declustered seismic chart used in the paper. The code is self-contained and embeds the six supplied events. It constructs:

1. a raw event cloud $\mathcal{D}_{\mathrm{raw}}$;
2. a mainshock sector $\mathcal{M}$ and aftershock sector $\mathcal{A}$;
3. a declustering projection $\Pi_M : \mathcal{D}_{\mathrm{raw}} \to \mathcal{M}$;
4. a local forward map $F(\theta)$ from a mainshock/window chart to normalized aftershock response coordinates;
5. a finite-difference Jacobian $J = dF/d\theta$;
6. an operational aftershock covariance C;
7. a Fisher recognition metric $J^T C^{-1} J$;
8. an observation null projector $P_N = I - U_r U_r^T$ onto $\ker(J^T)$;
9. a null-sector covariance $P_N C P_N$.

The purpose of the appendix is not to provide a seismic forecast model. It documents exactly how the CTMT words “chart”, “Jacobian”, “Fisher recognition”, and “null sector” are translated into a local seismic catalogue calculation.

References

[1] S. Khurram and P. Khalid, “Statistical analysis of earthquake catalogue and magnitude of completeness for Northern and Southwestern Pakistan,” *Acta Geodynamica et Geomaterialia*, 19(1), 23–34, 2022. DOI: 10.13168/AGG.2021.0040.

[2] F. J. Núñez-Cornú et al., “The Jalisco Seismic Accelerometric Telemetric Network (RESAJ),” *Seismological Research Letters.*

[3] C. Chauhan, “Earthquake dataset,” Kaggle dataset page, accessed for column definitions and catalogue context.